

CMPE 297-01 SP26

Computer Vision

Deep Learning for Computer Vision

CLASS LOGISTICS

Introductions



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Instructor



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The Plan for Today

- Course goals and logistics
- Introduction to Computer Vision

Information and Resources

- Syllabus published (partially)
- Office hours: Course website and Canvas available

Lectures and Office Hours

- Lectures
 - In-person
 - ENG DMH 348
- Office Hours
 - Walk-in or by appointment
 - Tue/Thu 1:00PM – 2:00 PM

Pre-Requisites

- CMPE 252 or CMPE 255 or CMPE 257
- Multi-variable calculus, linear algebra, statistics
- Some Prior knowledge of Machine Learning is required
 - Supervised/unsupervised learning, SGD, Calculus, Linear Algebra, Statistics
- Some Prior knowledge of Deep Learning is required
 - Backpropagation, Neural Nets, sequence models
- Tutorial on PyTorch, Python as related to CV
- Tutorial on OpenCV with Python

Coursework and Grading

- **Grade Breakdown**
- Quizzes: 10%
- Written Assignments: 10%
- Programming Assignments: 10%
- Midterm Exam: 25%
- Course Project: 20%
 - Proposal: 2%
 - Check-in Milestones: 3%
 - Project Presentation: 5%
 - Project Paper: 10%
- Final Exam: 25%

Coursework

- Written Assignment Homework
Solving theoretical exercise and problems
- Programming Assignment Homework
Implement different RL methods in PyTorch, Python, Gymnasium
- Final Project
 - Teams of 3 students will be created
 - Custom project: The team may propose its own topic
 - Default project – Instructor assigned end-to-end pipeline
 - Grouping students in teams at end of second week

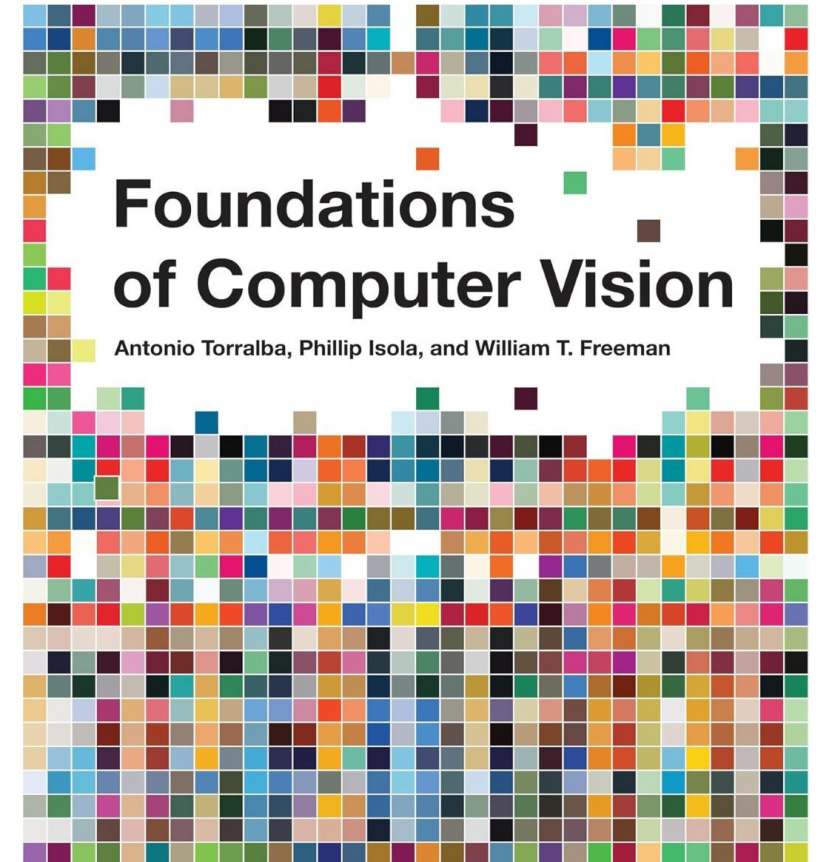
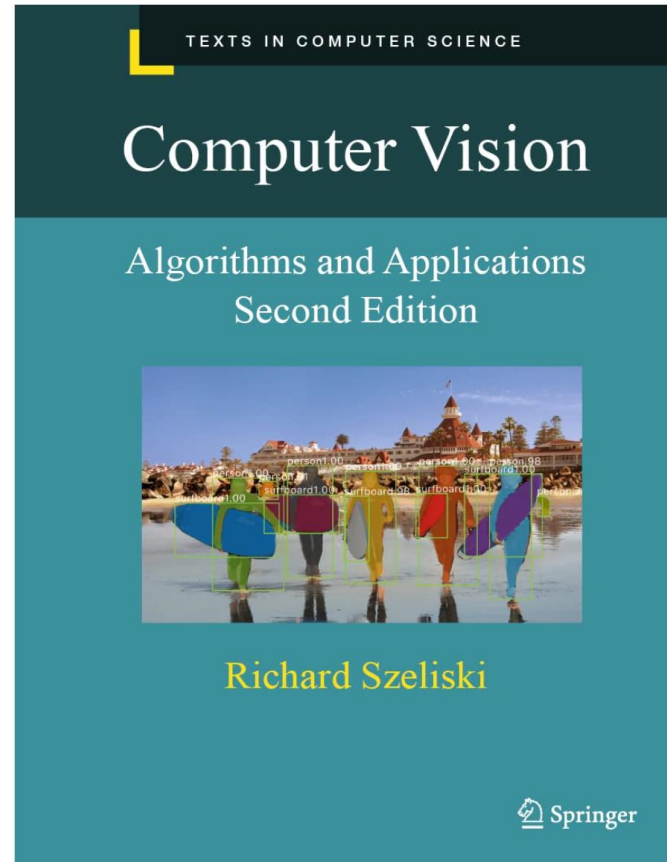
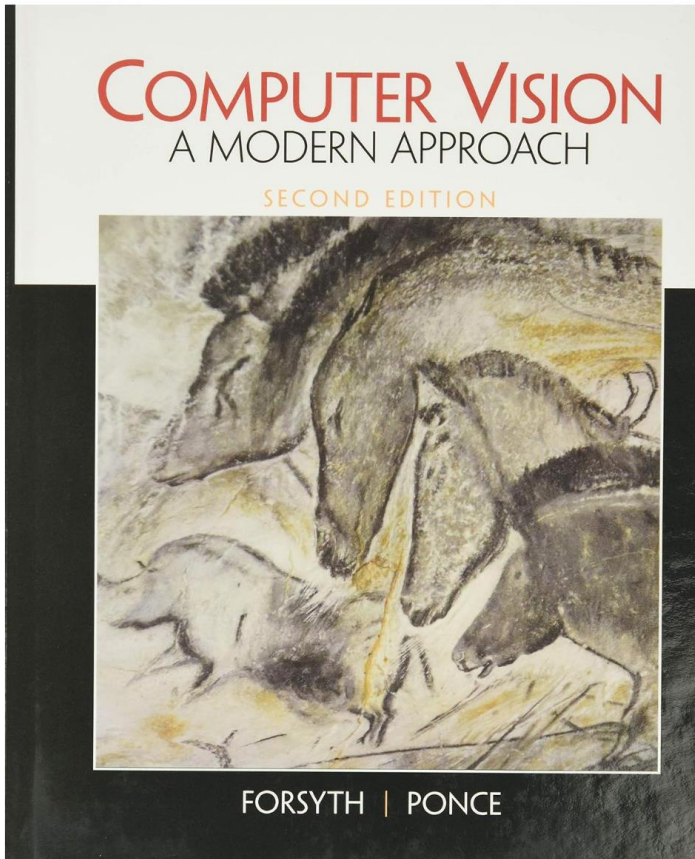
Course Exams

- Midterm Exam
- Final Exam
- Both exams have are in two sessions: One written and the other oral
- Exact date and time of both exams will be publish in the next update of the syllabus

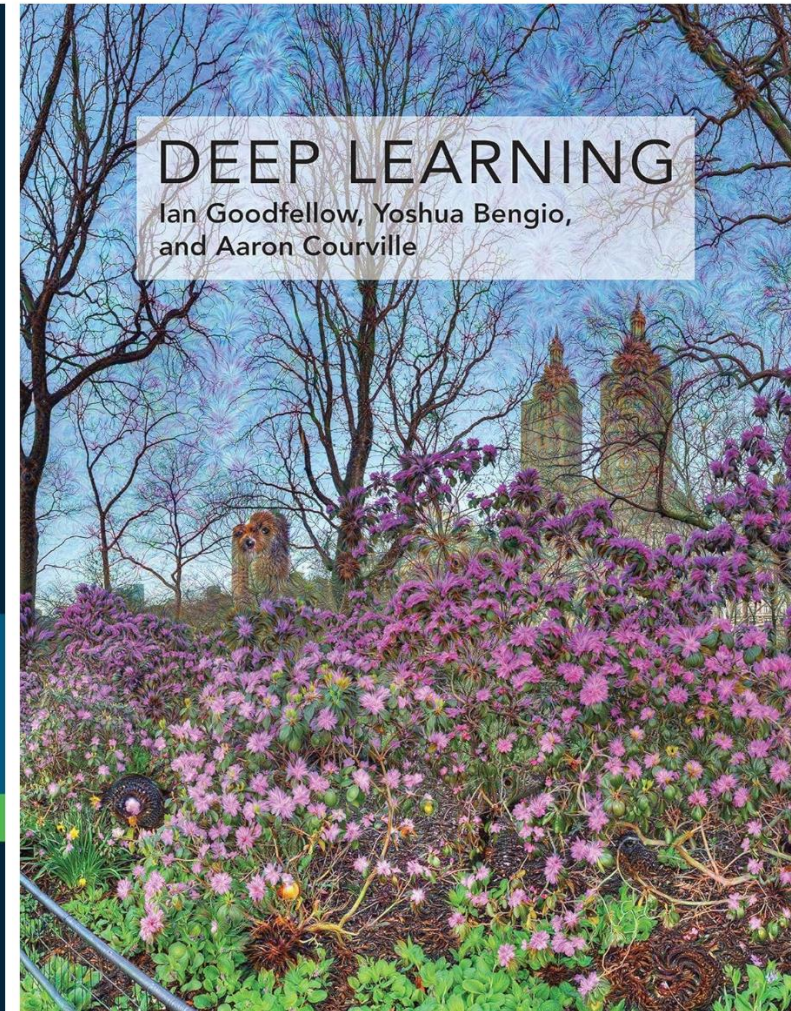
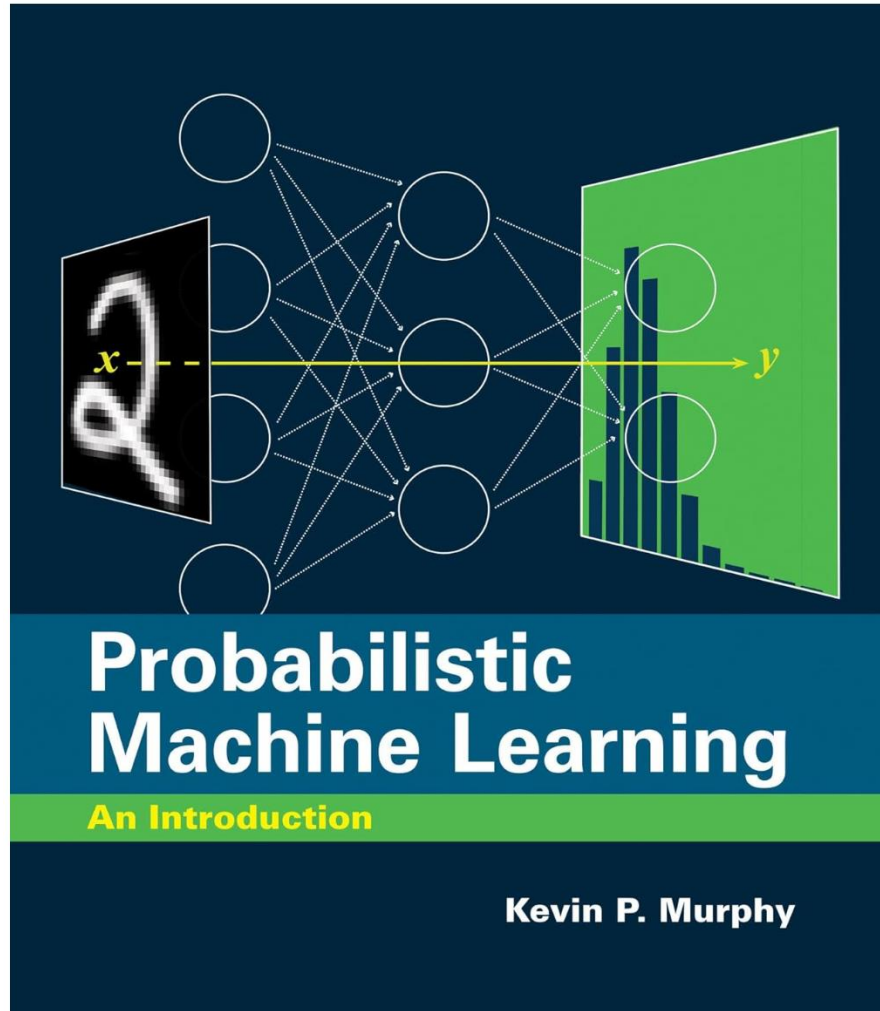
Course Policies

- The course official syllabus will be out next Monday
- Please make sure to read the course policies!

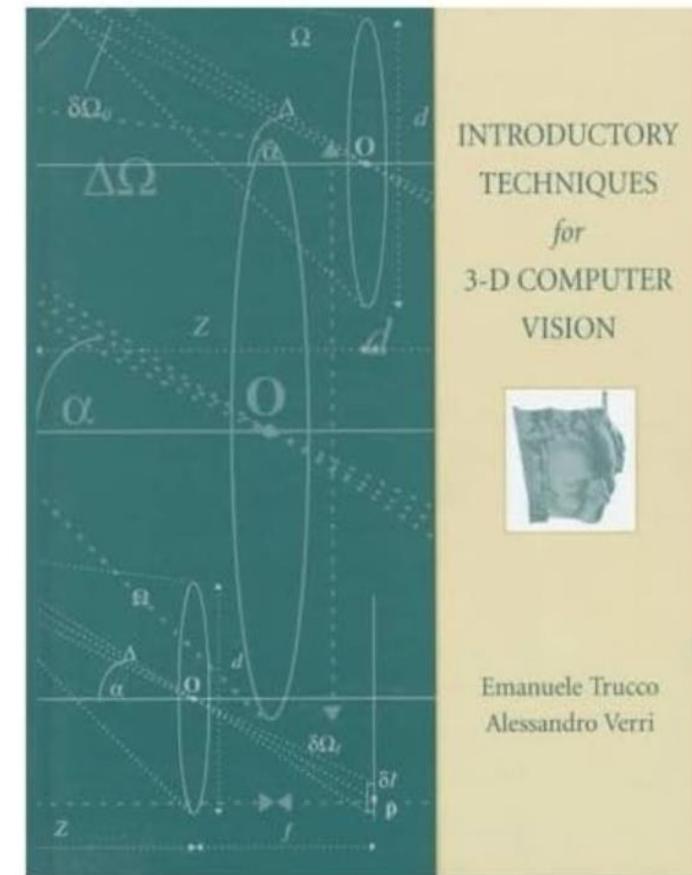
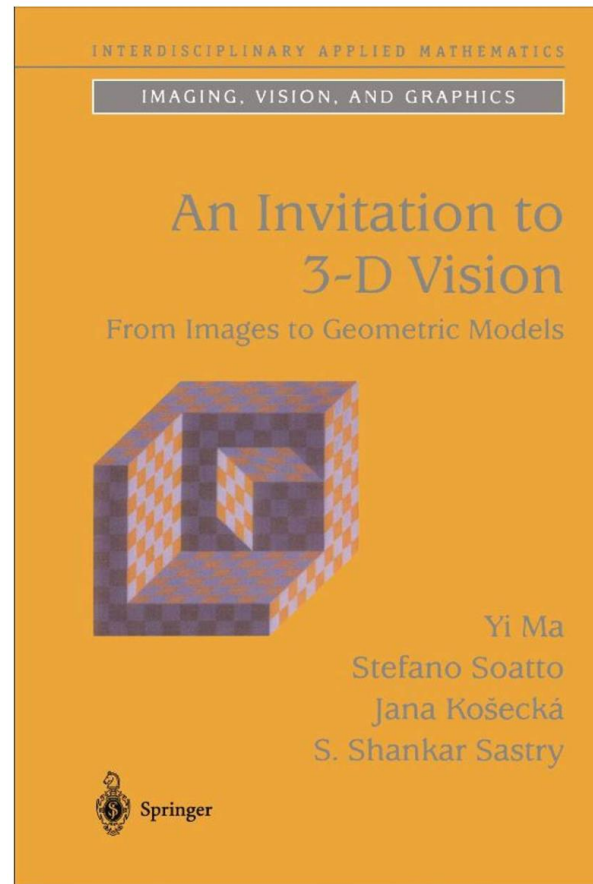
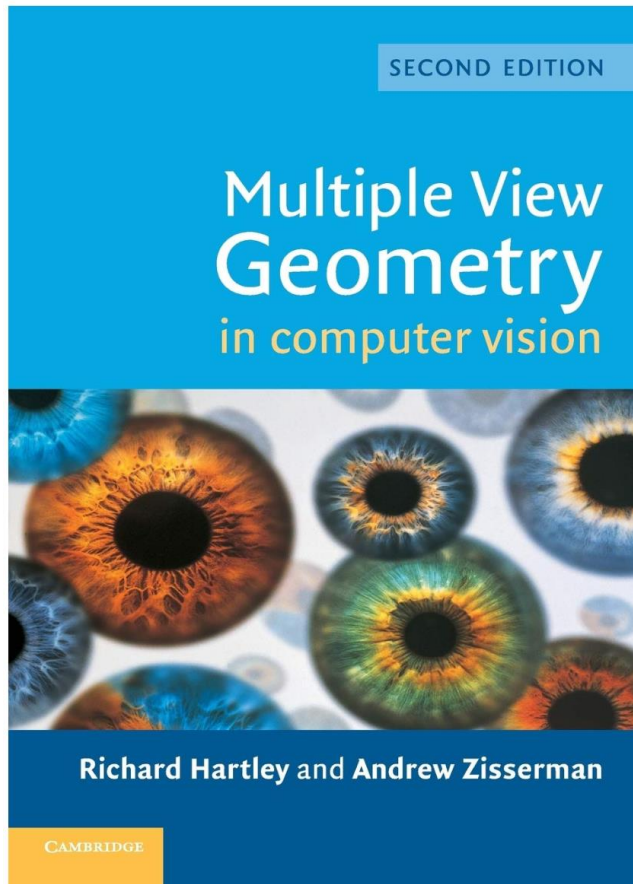
Computer Vision References



Deep Learning References



3-D Vision References



Looking Forward to a Great Semester!